Recitation 1 Solutions

Problem 1 - Vectors

A small online store records the number of orders received during 12 consecutive hours: 18, 24, 15, 31, 27, 22, 35, 19, 28, 40, 26, 33.

1. Create a vector called `orders` containing these values.

```
orders <- c(18, 24, 15, 31, 27, 22, 35, 19, 28, 40, 26, 33)

print(orders)
```

```
##  [1] 18 24 15 31 27 22 35 19 28 40 26 33
```

2. Use indexing to extract:

- the first four observations

```
orders[1:4]
```

```
## [1] 18 24 15 31
```

- the 3rd, 6th, 9th, and 12th observations

```
orders[c(3, 6, 9, 12)]
```

```
## [1] 15 22 28 33
```

- all observations except the first and last

```
orders[-c(1, 12)]
```

```
##  [1] 24 15 31 27 22 35 19 28 40 26
```

3. Create a new vector called `high_orders` containing only values from `orders` that are greater than 25.

```
high_orders <- orders[orders > 25]

print(high_orders)
```

```
## [1] 31 27 35 28 40 26 33
```

4. The store later discovers that the 4th recorded value should have been 29 rather than 31. Modify `orders` accordingly.

```r
orders[4] <- 29

print(orders)
```

```
##  [1] 18 24 15 29 27 22 35 19 28 40 26 33
```

5. Insert the value 21 immediately after the second observation and save the resulting vector as `order_updated`.

```r
orders_updated <- append(orders, 21, after = 2)

print(orders_updated)
```

```
##  [1] 18 24 21 15 29 27 22 35 19 28 40 26 33
```

6. Using a vectorized operation, create a vector called `adjusted_orders` in which every value of `ordered_updated` is increased by 10%.

```r
adjusted_orders <- orders_updated * 1.10

print(adjusted_orders)
```

```
##  [1] 19.8 26.4 23.1 16.5 31.9 29.7 24.2 38.5 20.9 30.8 44.0 28.6 36.3
```

7. Create the following vector without typing every element individually: 5, 10, 15, 20, 25, 30, 35, 40.

```r
targets <- seq(from = 5, to = 40, by = 5)

print(targets)
```

```
## [1]  5 10 15 20 25 30 35 40
```

Problem 2 - Lists

Suppose we want to store information about a statistical computing course. Create a list called `course_info` containing the following five elements: "Applied Statistical Computing", 35, the vector ["R", "Python"], the vector [88, 92, 79, 95, 84], another list containing "Fall 2026" and TRUE.

1. create `course_info`.

```r
course_info <- list("Applied Statistical Computing", 35, c("R", "Python"),

                    c(88, 92, 79, 95, 84), list("Fall 2026", TRUE))

print(course_info)
```

```
## [[1]]
## [1] "Applied Statistical Computing"
##
## [[2]]
## [1] 35
##
## [[3]]
## [1] "R"      "Python"
##
## [[4]]
## [1] 88 92 79 95 84
##
## [[5]]
## [[5]][[1]]
## [1] "Fall 2026"
##
## [[5]][[2]]
## [1] TRUE
```

2. Display the third element of `course info` using single ([]) and double ([[]]) brackets. Examine the outputs. What is the difference?

```
course_info[3]
```

```
## [[1]]
## [1] "R"      "Python"
```

```
course_info[[3]]
```

```
## [1] "R"      "Python"
```

Single brackets produce a list and double brackets produce a vector.

3. Extract `Python` directly from `course_info`.

```
course_info[[3]][2]
```

```
## [1] "Python"
```

4. Extract the value `95` directly from `course_info`.

```
course_info[[4]][4]
```

```
## [1] 95
```

5. Extract `Fall 2026` directly from `course_info`.

```r
course_info[[5]][[1]]
```

```
## [1] "Fall 2026"
```

6. Change the second element of `course_info` from 35 to 40.

```r
course_info[[2]] <- 40

print(course_info)
```

```
## [[1]]
## [1] "Applied Statistical Computing"
##
## [[2]]
## [1] 40
##
## [[3]]
## [1] "R"      "Python"
##
## [[4]]
## [1] 88 92 79 95 84
##
## [[5]]
## [[5]][[1]]
## [1] "Fall 2026"
##
## [[5]][[2]]
## [1] TRUE
```

7. Change the second programming language from "Python" to "SQL".

```r
course_info[[3]][2] <- "SQL"

print(course_info[[3]])
```

```
## [1] "R"   "SQL"
```

8. Add "Statistics Department" as a new sixth element of `course_info`.

```r
course_info[[6]] <- "Statistics Department"

print(course_info)
```

```
## [[1]]
## [1] "Applied Statistical Computing"
##
## [[2]]
## [1] 40
##
## [[3]]
```

```
## [1] "R"   "SQL"
##
## [[4]]
## [1] 88 92 79 95 84
##
## [[5]]
## [[5]][[1]]
## [1] "Fall 2026"
##
## [[5]][[2]]
## [1] TRUE
##
##
## [[6]]
## [1] "Statistics Department"
```

9. Remove the fourth element of the list.

```
course_info[[4]] <- NULL

print(course_info)
```

```
## [[1]]
## [1] "Applied Statistical Computing"
##
## [[2]]
## [1] 40
##
## [[3]]
## [1] "R"   "SQL"
##
## [[4]]
## [[4]][[1]]
## [1] "Fall 2026"
##
## [[4]][[2]]
## [1] TRUE
##
##
## [[5]]
## [1] "Statistics Department"
```

10. Create another list called `instructor_info` containing "Dr. Smith" and "Columbia University". Combine `course_info` and `instructor_info` using `c()` and save the result as `combined1`. Next, combine `course_info` and `instructor_info` using `list()` and save the result as `combined2`.Compare `combined1` and `combined2`. How many elements does each contain?

```
instructor_info <- list("Dr. Smith", "Columbia University")

combined1 <- c(course_info, instructor_info)

print(combined1)
```

```
## [[1]]
## [1] "Applied Statistical Computing"
##
## [[2]]
## [1] 40
##
## [[3]]
## [1] "R"   "SQL"
##
## [[4]]
## [[4]][[1]]
## [1] "Fall 2026"
##
## [[4]][[2]]
## [1] TRUE
##
##
## [[5]]
## [1] "Statistics Department"
##
## [[6]]
## [1] "Dr. Smith"
##
## [[7]]
## [1] "Columbia University"
```

```r
combined2 <- list(course_info, instructor_info)

print(combined2)
```

```
## [[1]]
## [[1]][[1]]
## [1] "Applied Statistical Computing"
##
## [[1]][[2]]
## [1] 40
##
## [[1]][[3]]
## [1] "R"   "SQL"
##
## [[1]][[4]]
## [[1]][[4]][[1]]
## [1] "Fall 2026"
##
## [[1]][[4]][[2]]
## [1] TRUE
##
##
## [[1]][[5]]
## [1] "Statistics Department"
##
##
## [[2]]
## [[2]][[1]]
```

```
## [1] "Dr. Smith"
##
## [[2]][[2]]
## [1] "Columbia University"
```

Problem 3 - Matrices

The following values represent quiz scores for 4 students on 3 quizzes:

- 82, 75, 91;
- 68, 84, 79;
- 95, 88, 93;
- 76, 81, 87.

1. Create a matrix called `scores` with 4 rows and 3 columns, arranging the values by row.

```
scores <- matrix( c(82, 75, 91, 68, 84, 79, 95, 88, 93, 76, 81, 87), nrow = 4,

                  byrow = TRUE)

print(scores)
```

```
##      [,1] [,2] [,3]
## [1,]   82   75   91
## [2,]   68   84   79
## [3,]   95   88   93
## [4,]   76   81   87
```

2. Extract:

- Student 3's Quiz 2 score;

```
scores[3, 2]
```

```
## [1] 88
```

- all scores for Student 2;

```
scores[2, ]
```

```
## [1] 68 84 79
```

- all Quiz 3 scores;

```
scores[, 3]
```

```
## [1] 91 79 93 87
```

- the scores for Students 1 and 4 on Quizzes 1 and 3.

```r
scores[c(1, 4), c(1, 3)]
```

```
##      [,1] [,2]
## [1,]   82   91
## [2,]   76   87
```

3. Student 2's Quiz 1 score was entered incorrectly. Replace 68 with 73.

```r
scores[2, 1] <- 73

print(scores)
```

```
##      [,1] [,2] [,3]
## [1,]   82   75   91
## [2,]   73   84   79
## [3,]   95   88   93
## [4,]   76   81   87
```

4. Add a fourth quiz using the scores 90, 85, 96, 82.

```r
scores <- cbind(scores, c(90, 85, 96, 82))

print(scores)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]   82   75   91   90
## [2,]   73   84   79   85
## [3,]   95   88   93   96
## [4,]   76   81   87   82
```

5. A fifth student joins the dataset with scores 80, 77, 89. Add this student as a new row to `scores` and call the resulting matrix `scores2`.

```r
scores2 <- rbind(scores, c(80, 77, 89, 91))

print(scores2)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]   82   75   91   90
## [2,]   73   84   79   85
## [3,]   95   88   93   96
## [4,]   76   81   87   82
## [5,]   80   77   89   91
```

6. Create a new matrix called `bonus_scores` by adding 3 points to every score in `scores2`. Do this using a vectorized matrix operation.

```r
bonus_scores <- scores2 + 3

print(bonus_scores)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]   85   78   94   93
## [2,]   76   87   82   88
## [3,]   98   91   96   99
## [4,]   79   84   90   85
## [5,]   83   80   92   94
```

7. In `bonus_scores`, replace every score greater than 95 with 100.

```r
bonus_scores[bonus_scores > 95] <- 100

print(bonus_scores)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]   85   78   94   93
## [2,]   76   87   82   88
## [3,]  100   91  100  100
## [4,]   79   84   90   85
## [5,]   83   80   92   94
```

8. Extract only those elements of `bonus_scores` that are at least 90.

```r
bonus_scores[bonus_scores >= 90]
```

```
## [1] 100  91  94 100  90  92  93 100  94
```

9. Remove Student 2 and Quiz 1 from `bonus_scores` and save the resulting matrix as `reduced_scores`.

```r
reduced_scores <- bonus_scores[-2, -1]

print(reduced_scores)
```

```
##      [,1] [,2] [,3]
## [1,]   78   94   93
## [2,]   91  100  100
## [3,]   84   90   85
## [4,]   80   92   94
```

Problem 4 - Data Frames

A company collected information about eight employees:

- Employee - [“Anna”, “Ben”, “Carla”, “David”, “Emma”, “Frank”, “Grace”, “Henry”]

- Department - [“Analytics”, “Marketing”, “Analytics”, “Finance”, “Marketing”, “Analytics”, “Finance”, “Analytics”]
- Experience - [2, 7, 5, 9, 3, 6, 8, 4]
- Salary - [68000, 85000, 79000, 105000, 72000, 92000, 99000, 76000]
- Remote - [TRUE, FALSE, TRUE, FALSE, TRUE, FALSE, TRUE, TRUE]

1. Use these vectors to create a data frame called `employees`.

```
Employee <- c("Anna", "Ben", "Carla", "David", "Emma", "Frank", "Grace",

              "Henry")

Department <- c("Analytics", "Marketing", "Analytics", "Finance", "Marketing",

                "Analytics", "Finance", "Analytics")

Experience <- c(2, 7, 5, 9, 3, 6, 8, 4)

Salary <- c(68000, 85000, 79000, 105000, 72000, 92000, 99000, 76000)

Remote <- c(TRUE, FALSE, TRUE, FALSE, TRUE, FALSE, TRUE, TRUE)


employees <- data.frame(Employee, Department, Experience, Salary, Remote)

print(employees)
```

```
##   Employee Department Experience Salary Remote
## 1     Anna  Analytics          2  68000   TRUE
## 2      Ben  Marketing          7  85000  FALSE
## 3    Carla  Analytics          5  79000   TRUE
## 4    David    Finance          9 105000  FALSE
## 5     Emma  Marketing          3  72000   TRUE
## 6    Frank  Analytics          6  92000  FALSE
## 7    Grace    Finance          8  99000   TRUE
## 8    Henry  Analytics          4  76000   TRUE
```

2. Use appropriate R commands to examine:

- the structure of `employees`;

```
str(employees)
```

```
## 'data.frame':    8 obs. of  5 variables:
##  $ Employee  : chr  "Anna" "Ben" "Carla" "David" ...
##  $ Department: chr  "Analytics" "Marketing" "Analytics" "Finance" ...
##  $ Experience: num  2 7 5 9 3 6 8 4
##  $ Salary    : num  68000 85000 79000 105000 72000 92000 99000 76000
##  $ Remote    : logi  TRUE FALSE TRUE FALSE TRUE FALSE ...
```

- a summary of its variables;

```r
summary(employees)
```

```
##    Employee          Department          Experience       Salary
##  Length:8           Length:8           Min.   :2.00   Min.   : 68000
##  Class :character   Class :character   1st Qu.:3.75   1st Qu.: 75000
##  Mode  :character   Mode  :character   Median :5.50   Median : 82000
##                                        Mean   :5.50   Mean   : 84500
##                                        3rd Qu.:7.25   3rd Qu.: 93750
##                                        Max.   :9.00   Max.   :105000
##    Remote
##  Mode :logical
##  FALSE:3
##  TRUE :5
##
##
##
```

- the first three rows;

```r
head(employees, 3)
```

```
##   Employee Department Experience Salary Remote
## 1     Anna  Analytics          2  68000   TRUE
## 2      Ben  Marketing          7  85000  FALSE
## 3    Carla  Analytics          5  79000   TRUE
```

- its dimensions.

```r
dim(employees)
```

```
## [1] 8 5
```

3. Extract the `Salary` variable in three different ways so that:

- one result is a data frame;

```r
employees["Salary"]
```

```
##   Salary
## 1  68000
## 2  85000
## 3  79000
## 4 105000
## 5  72000
## 6  92000
## 7  99000
## 8  76000
```

- the other two results are vectors;

```r
employees[["Salary"]]
```

```
## [1]  68000  85000  79000 105000  72000  92000  99000  76000
```

```r
#or

employees$Salary
```

```
## [1]  68000  85000  79000 105000  72000  92000  99000  76000
```

4. Extract only the `Employee`, `Department`, and `Salary` columns.

```r
employees[, c("Employee", "Department", "Salary")]
```

```
##   Employee Department Salary
## 1     Anna  Analytics  68000
## 2      Ben  Marketing  85000
## 3    Carla  Analytics  79000
## 4    David    Finance 105000
## 5     Emma  Marketing  72000
## 6    Frank  Analytics  92000
## 7    Grace    Finance  99000
## 8    Henry  Analytics  76000
```

5. Extract rows 2 through 6 and only the `Employee` and `Experience` columns.

```r
employees[2:6, c("Employee", "Experience")]
```

```
##   Employee Experience
## 2      Ben          7
## 3    Carla          5
## 4    David          9
## 5     Emma          3
## 6    Frank          6
```

6. Extract all employees who have at least 5 years of experience.

```r
employees[employees$Experience >= 5, ]
```

```
##   Employee Department Experience Salary Remote
## 2      Ben  Marketing          7  85000  FALSE
## 3    Carla  Analytics          5  79000   TRUE
## 4    David    Finance          9 105000  FALSE
## 6    Frank  Analytics          6  92000  FALSE
## 7    Grace    Finance          8  99000   TRUE
```

7. Add a new variable called `Salary_K` that contains salaries expressed in thousands of dollars. For example, 68000 becomes 68.

```r
employees$Salary_K <- employees$Salary / 1000

print(employees)
```

```
##   Employee Department Experience Salary Remote Salary_K
## 1     Anna  Analytics          2  68000   TRUE       68
## 2      Ben  Marketing          7  85000  FALSE       85
## 3    Carla  Analytics          5  79000   TRUE       79
## 4    David    Finance          9 105000  FALSE      105
## 5     Emma  Marketing          3  72000   TRUE       72
## 6    Frank  Analytics          6  92000  FALSE       92
## 7    Grace    Finance          8  99000   TRUE       99
## 8    Henry  Analytics          4  76000   TRUE       76
```

8. Add another variable called `Senior` that takes the value `TRUE` when `Experience` is at least 6 years and `FALSE` otherwise.

```r
employees$Senior <- employees$Experience >= 6

print(employees)
```

```
##   Employee Department Experience Salary Remote Salary_K Senior
## 1     Anna  Analytics          2  68000   TRUE       68  FALSE
## 2      Ben  Marketing          7  85000  FALSE       85   TRUE
## 3    Carla  Analytics          5  79000   TRUE       79  FALSE
## 4    David    Finance          9 105000  FALSE      105   TRUE
## 5     Emma  Marketing          3  72000   TRUE       72  FALSE
## 6    Frank  Analytics          6  92000  FALSE       92   TRUE
## 7    Grace    Finance          8  99000   TRUE       99   TRUE
## 8    Henry  Analytics          4  76000   TRUE       76  FALSE
```

9. Carla's salary should be 82,000 instead of 79,000. Modify the appropriate element of the data frame.

```r
employees[3, "Salary"] <- 82000
```

10. Add the following employee as a new row: Employee - Irene; Department - Marketing; Experience - 10; Salary - 110000; Remote - FALSE.

```r
Irene <- data.frame(Employee = "Irene", Department = "Marketing",

                    Experience = 10, Salary = 110000, Remote = FALSE,

                    Salary_K = 110, Senior = TRUE)

employees <- rbind(employees, Irene)

print(employees)
```

```
##   Employee Department Experience Salary Remote Salary_K Senior
## 1     Anna  Analytics          2  68000   TRUE       68  FALSE
```

```
## 2      Ben  Marketing          7  85000  FALSE       85   TRUE
## 3    Carla  Analytics          5  82000   TRUE       79  FALSE
## 4    David    Finance          9 105000  FALSE      105   TRUE
## 5     Emma  Marketing          3  72000   TRUE       72  FALSE
## 6    Frank  Analytics          6  92000  FALSE       92   TRUE
## 7    Grace    Finance          8  99000   TRUE       99   TRUE
## 8    Henry  Analytics          4  76000   TRUE       76  FALSE
## 9    Irene  Marketing         10 110000  FALSE      110   TRUE
```